

## ANTIBACTERIAL STUDIES OF SULFACETAMIDE OPHTHALMIC SOLUTIONS AND SUSPENSIONS

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### ABSTRACT:

During present study sulfacetamide alone as well as its combination were evaluated against clinical isolates of *Escherichia coli* and *Staphylococcus aureus*.

For this purpose fifty clinical isolates of these organisms were collected from different pathological laboratories and antibacterial activity was carried out using ophthalmic solutions and suspensions.

40% clinical isolates of *Escherichia coli* were found to be sensitive to four brands of ophthalmic solutions, 56% isolates showed sensitivity against one brand where as 4% isolates were resistant to all preparations.

When sulfacetamide was used in combination. 20% clinical isolates of *Escherichia coli* were completely sensitive to different brands used while 80% isolates were found to be resistant to all the brands.

Antibacterial activity of sulfacetamide alone indicated that 92% clinical isolates of *Staphylococcus aureus* were sensitive to all the brands used and only 8% isolates were resistant.

Sensitivity of *Staphylococcus aureus* against different combinations of sulfacetamide indicated that 68% isolates were sensitive to all brands while only 32% were resistant to different combinations.

### INTRODUCTION

The first sulfonamide of clinical importance, "Prontosil rubrum", was synthesized in Germany in 1932, and first used as a chemotherapeutic agent in 1935, initiating a new era in the treatment of infections.

It was soon shown that the therapeutic action of this compound depended on its breakdown in the body into an inactive dye and an antibacterial substance called sulfanilamide (p- amino- benzene sulfonamide) which had been synthesized in 1908.

Subsequently, by manipulating chemical side- chains, several thousand sulfonamides were synthesized and some of these have clinical uses other than for chemotherapy, such as the oral hypoglycemic agent, tolbutamide.

The therapeutic role of sulfonamide alone

is steadily diminishing. In combination with other agents such as trimethoprim or pyrimethamine, however, a number of sulfonamides remain clinically relevant.

A classification of clinically useful sulfonamide is based on their absorption and excretion pattern. They may be divided into several groups:

**Short-acting sulfonamides**→readily absorbed from G.I tract and are rapidly excreted.

**Medium-acting sulfonamides**→well absorbed from G.I tract, more slowly excreted so that 12-hourly administrations is adequate.

**Long-acting sulfonamides**→rapidly absorbed from G.I tract, slowly excreted, administered once daily.

**Ultra-long-acting sulfonamide**→therapeutic

serum and urine levels persist for upto 7 days after a single oral dose of 1-2 g.

**Poorly absorbed sulfonamides** → most of the orally administered dose is excreted unchanged in the feces and their action is confined to gut. These drugs are rarely used nowadays (Kucers *et al.*, 1997).

It is difficult to formulate dosage schedules for patients with renal failure as individual sulfonamide are excreted by different renal mechanisms and at different rates (Fisher, 1972).

Adam (1970) and Adam *et al* (1973) in studies of sulfadiazine, sulfadiazine and sulfamethazole found that, with moderate doses, both the free drugs and their conjugated derivatives accumulated in the plasma of uremic patients. In some serum levels appeared to cause side-effects. Further more, the high concentration of free sulfonamides were low in uremic patients. These authors concluded that sulfonamides are unlikely to be useful for treatment of urinary tract infections in patients with serum creatinine level exceeding 5 mg% (0.4 mmol/liter), and that for patients with lesser degree of renal failure, a reduced sulfonamide dose may be required to avoid toxicity.

For treatment of systemic infections, sulfonamides can probably be safely used in patients with renal disease, but regular estimations of the serum concentration of both free and acetylated sulfonamide seem advisable.

The various sulfonamides are handled by the kidney in different ways, and the rate of acetylation of sulfonamides in the body varies in individual patients.

There are now very few specific indications for sulfonamides, because of range of available antibiotics. Emergence of the wide sulfonamide-resistance among many bacterial species has made these drugs unsuitable for the treatment of many diseases, for which they

had been used successfully in the past. The combination of sulfonamides with trimethoprim, however, is used for many diseases and the combination of sulfadiazine and pyrimethamine is often used for toxoplasma encephalitis.

Sulfonamides alone are often successful for the treatment of acute uncomplicated urinary tract infections due to susceptible pathogens. However, despite their generally low cost, their use for this indication is diminishing due to increasing antibiotic resistance and the availability of effective agents with lower rates of toxicity.

Sulfonamides with medium duration of action (e.g. sulfamethoxazole) are probably preferable to short-acting sulfonamides (e.g. sulfamethizole) although in at least one trial there was no therapeutic difference between these two agents (Bergaant and Skjerven, 1979).

Similar to other antibiotic agents, single-dose sulfonamide regimens have proved effective for the treatment of acute uncomplicated urinary tract infections (Kallénius, 1979). Sulfonamides are not usually used for chronic and recurrent infections or in hospital-acquired, infections.

These drugs have been used for treatment of *Strep. pyogenes* throat infections (Alban, 1965), but penicillin G is the drug of choice for these infections and safe, effective alternative antibiotics are available for penicillin- allergic patients. Ampicillin is usually the preferred treatment of otitis media in children, but combinations of a sulfonamide with erythromycin or phenoxymethyl penicillin can be used (Paradise, 1980 and Bluestone, 1982).

Sulfonamides have been used for treatment of *Pneumococcal pneumonia* and other respiratory tract bacterial infections, but safer and more effective drugs are available. Sulfonamides are useful for the treatment of some mycobacterial infections.

**Table-1**

| Code no. | Contents                                                                         | Dosage form |
|----------|----------------------------------------------------------------------------------|-------------|
| 01(a)    | sulfacetamide Na 10%                                                             | solution    |
| 01(b)    | sulfacetamide Na 20%                                                             | solution    |
| 01(c)    | sulfacetamide Na 30%                                                             | solution    |
| 02(a)    | sulfacetamide Na 10%                                                             | solution    |
| 02(b)    | sulfacetamide Na 20%                                                             | solution    |
| 02(c)    | sulfacetamide Na 30%                                                             | solution    |
| 03(a)    | sulfacetamide Na 10%                                                             | solution    |
| 03(b)    | sulfacetamide Na 20%                                                             | solution    |
| 04(a)    | sulfacetamide Na 10%                                                             | solution    |
| 04(b)    | sulfacetamide Na 30%                                                             | solution    |
| 05       | sulfacetamide Na 10%<br>prednisolone acetate 0.25%                               | solution    |
| 06       | sulfacetamide Na 10% +<br>prednisolone acetate 0.25%                             | solution    |
| 07       | sulfacetamide Na 10% +<br>prednisolone acetate 0.25%                             | suspension  |
| 08       | sulfacetamide Na 10% +<br>prednisolone acetate 0.25%                             | suspension  |
| 09       | sulfacetamide Na 10% +<br>prednisolone acetate 0.2% +<br>phenylephrine HCl 0.12% | solution    |
| 10       | sulfacetamide Na 10% +<br>prednisolone acetate 0.2% +<br>phenylephrine HCl 0.12% | solution    |
| 11       | sulfacetamide Na 10% +<br>dexamethasone 0.1%                                     | solution    |

***Ophthalmic Solution:***

This is by far the most common means of administering a drug to the eye by definition, all ingredients are completely in solution, uniformity is not a problem and there is little physical interference with vision. The principal disadvantage of solutions is the relatively brief contact time between the medication and absorbing surfaces. Contact time may be increased to some extent by the inclusion of a viscosity increasing agent such as methylcellulose. A viscosity in the range of 15 to 25 cps is considered optimum for drug retention and visual comfort.

***Ophthalmic Suspension:***

Suspensions are dispersions of finely divided, relatively insoluble drug substances in an aqueous vehicle containing suitable suspending and dispersing agents. The vehicle is among other things, a saturated solution of the drug substances. Because of a tendency of particles to be retained in the cul-de-sac, the contact time and duration of action of a suspension probably exceeds that of a solution. The drug is absorbed from solution and the solution concentration is replenished from retained particles.

**Table 2A**  
Different Percentages of Sulfacetamide (Alone) of Brand 01 and 02

| S.No. of clinical isolates of <i>Escherichia coli</i> | Zone of inhibition (mm) Brand 01 |                  |                 | Zone of inhibition (mm) Brand 02 |                  |                  |
|-------------------------------------------------------|----------------------------------|------------------|-----------------|----------------------------------|------------------|------------------|
|                                                       | 01(a)<br>sol 10%                 | 01(b)<br>sol 20% | 01(c)<br>sol 30 | 02(a)<br>sol 10%                 | 02(b)<br>sol 20% | 02(c)<br>sol 30% |
| 1                                                     | 20.0                             | 20.0             | 30.0            | 20.0                             | 25.0             | 25.0             |
| 2                                                     | R                                | R.               | R               | R                                | R                | R                |
| 3                                                     | R                                | R                | R               | R                                | R                | R                |
| 4                                                     | R                                | R                | R               | R                                | R                | R                |
| 5                                                     | R                                | R                | R               | R                                | R                | R                |
| 6                                                     | 23.0                             | 24.0             | 26.0            | 25.0                             | 25.0             | 33.0             |
| 7                                                     | R                                | R                | R               | R                                | R                | R                |
| 8                                                     | 20.0                             | 20.0             | 25.0            | 10.0                             | 12.0             | 13.0             |
| 9                                                     | R                                | R                | R               | R                                | R                | R                |
| 10                                                    | 11.0                             | 13.0             | 15.0            | 10.0                             | 15.0             | 15.0             |
| 11                                                    | R                                | R                | R               | R                                | R                | R                |
| 12                                                    | R                                | R                | R               | R                                | R                | R                |
| 13                                                    | R                                | R                | R               | R                                | R                | R                |
| 14                                                    | R                                | R                | R               | 35.0                             | 36.0             | 38.0             |
| 15                                                    | R                                | R                | R               | R                                | R                | R                |
| 16                                                    | R                                | R                | R               | R                                | R                | R                |
| 17                                                    | 10.0                             | 12.0             | 15.0            | 13.0                             | 15.0             | 16.0             |
| 18                                                    | 12.0                             | 14.0             | 14.0            | 11.0                             | 13.0             | 16.0             |
| 19                                                    | 12.0                             | 14.0             | 14.0            | 10.0                             | 10.0             | 13.0             |
| 20                                                    | R                                | R                | R               | R                                | R                | R                |
| 21                                                    | R                                | R                | R               | R                                | R                | R                |
| 22                                                    | 15.0                             | 15.0             | 16.0            | 15.0                             | 15.0             | 15.0             |
| 23                                                    | R                                | R                | R               | R                                | R                | R                |
| 24                                                    | 10.0                             | 10.0             | 12.0            | 13.0                             | 14.0             | 15.0             |
| 25                                                    | R                                | R                | R               | R                                | R                | R                |

Each of these actions is a function of particle size, with solubility rate being favored by smaller size and retention favored by a larger size, thus, optimum activity should result from an optimum particle size (Alfanso *et al.*, 1995).

The objective of the preset work is to evaluate the antibacterial activity of different ophthalmic solution and suspension of sulfacetamide sodium alone and in combinations available in market.

## ANTIMICROBIAL TESTING

### *Assay Medium:*

Mueller-Hinton Agar (MHA) was used for determining the sensitivity of bacteria by single disk diffusion method Kirby Bauer against different antimicrobial agents (Mueller-Hinton, 1941; Bauer *et al.*, 1966; Ericsson and Sherris, 1971).

**Table 2B**  
Different Percentages of Sulfacetamide (Alone) of Brand 03 and 04

| S. No. of clinical isolates of <i>Escherichia coli</i> | Zone of inhibition (mm) Brand 03 |                  | Zone of inhibition (mm) Brand 04 |                  |
|--------------------------------------------------------|----------------------------------|------------------|----------------------------------|------------------|
|                                                        | 03(a)<br>sol 10%                 | 03(b)<br>sol 20% | 04(a)<br>sol 10%                 | 04(b)<br>sol 20% |
| 1                                                      | 25.0                             | 30.0             | 30.0                             | 30.0             |
| 2                                                      | R                                | R                | 10.0                             | 10.0             |
| 3                                                      | R                                | R                | 10.0                             | 13.0             |
| 4                                                      | R                                | R                | 13.0                             | 16.0             |
| 5                                                      | R                                | R                | 10.0                             | 19.0             |
| 6                                                      | 25.0                             | 27.0             | 25.0                             | 30.0             |
| 7                                                      | R                                | R                | 14.0                             | 15.0             |
| 8                                                      | 20.0                             | 25.0             | 13.0                             | 14.0             |
| 9                                                      | R                                | R                | 15.0                             | 17.0             |
| 10                                                     | 17.0                             | 19.0             | 18.0                             | 20.0             |
| 11                                                     | R                                | R                | 13.0                             | 15.0             |
| 12                                                     | R                                | R                | 15.0                             | 18.0             |
| 13                                                     | R                                | R                | 15.0                             | 30.0             |
| 14                                                     | 30.0                             | 40.0             | 40.0                             | 45.0             |
| 15                                                     | R                                | R                | 10.0                             | 10.0             |
| 16                                                     | R                                | R                | 10.0                             | 10.0             |
| 17                                                     | R                                | R                | 10.0                             | 10.0             |
| 18                                                     | R                                | R                | 15.0                             | 15.0             |
| 19                                                     | 15.0                             | 18.0             | 14.0                             | 15.0             |
| 20                                                     | R                                | R                | 15.0                             | 20.0             |
| 21                                                     | R                                | R                | 10.0                             | 12.0             |
| 22                                                     | 17.0                             | 20.0             | 20.0                             | 22.0             |
| 23                                                     | R                                | R                | 10.0                             | 15.0             |
| 24                                                     | 14.0                             | 17.0             | 10.0                             | 10.0             |
| 25                                                     | R                                | R                | R                                | R                |

**a) Composition:**

|                    |           |
|--------------------|-----------|
| Meat infusion      | 5.0g      |
| Casein hydrolysate | 17.5 g    |
| Starch             | 1.5 g     |
| Agar-agar          | 13.5 g    |
| Distilled water    | 1.0 lit   |
| pH                 | 7.3 V 0.1 |

**b) Preparation:**

36.5 g of dehydrated Mueller-Hinton Agar (MHA) was suspended in 1 liter of distilled water till dissolved completely, autoclaved for 15 minutes at 121°C 15 lbs. Pressure. It was cooled at 45-50°C in a water bath before

using.

**Incubation:**

Within 15 minutes of placing the disks on the agar, the plates were inverted and incubated at 35-37°C for 16-18 hours. After incubation the diameter of clear zones around the sulfacetamide Na discs were measured using vernier calipers.

**Interpretation:**

The zone diameter of each antimicrobial agent was then interpreted as sensitive or resistant as shown in Tables 2-5.

**Table 3**  
Code # of Different Brands used in Combination  
(Sulfacetamide Na 10% + Prednisolone Acetate 0.25%, 12% and 0.1%)

| S. No of clinical isolates of <i>Escherichia coli</i> | Zone of inhibition (mm) |        |        |        |        |        |        |
|-------------------------------------------------------|-------------------------|--------|--------|--------|--------|--------|--------|
|                                                       | 0.25%                   |        |        |        | 0.12%  |        | 0.1%   |
|                                                       | 05 sol                  | 06 sol | 07 sol | 08 sol | 09 sol | 10 sol | 11 sol |
| 1                                                     | 15.0                    | 25.0   | 25.0   | 12.0   | 20.0   | 24.0   | 20.0   |
| 2                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 3                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 4                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 5                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 6                                                     | 25.0                    | 24.0   | 10.0   | 28.0   | 28.0   | 20.0   | 25.0   |
| 7                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 8                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 9                                                     | R                       | R      | R      | R      | R      | R      | R      |
| 10                                                    | 15.0                    | 10.0   | 10.0   | 16.0   | 14.0   | 15.0   | 15.0   |
| 11                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 12                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 13                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 14                                                    | 35.0                    | 35.0   | 35.0   | 40.0   | 37.0   | 40.0   | 35.0   |
| 15                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 16                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 17                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 18                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 19                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 20                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 21                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 22                                                    | 15.0                    | 10.0   | 10.0   | 15.0   | 15.0   | 16.0   | 19.0   |
| 23                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 24                                                    | R                       | R      | R      | R      | R      | R      | R      |
| 25                                                    | R                       | R      | R      | R      | R      | R      | R      |

## RESULT

During the present study, sulfacetamide alone as well as its combinations available in the market were evaluated against clinical isolates of *Escherichia coli* and *Staphylococcus aureus* Tables 2-5.

For this purpose fifty clinical isolates of these organisms were collected from different pathological laboratories and hospitals.

Sulfacetamide ophthalmic solutions and suspensions were purchased from local market Table-1.

## DISCUSSION

Sulfacetamide originally had a wide range of activity, but this range has now been restricted by acquired bacterial resistance and it is not true only for sulfacetamide rather all the other antimicrobials are becoming

**Table 4**  
Different Percentages of Sulfacetamide (Alone) of Brand 01, 02, 03 and 04

| S. No. of clinical isolates of <i>Staphylococcus aureus</i> | Zone of inhibition (mm) |      |      |          |      |      |          |      |          |      |
|-------------------------------------------------------------|-------------------------|------|------|----------|------|------|----------|------|----------|------|
|                                                             | Brand 01                |      |      | Brand 02 |      |      | Brand 03 |      | Brand 04 |      |
| 1                                                           | 30.0                    | 30.0 | 40.0 | 30.0     | 32.0 | 34.0 | 40.0     | 48.0 | 40.0     | 44.0 |
| 1                                                           | 12.0                    | 20.0 | 20.0 | 10.0     | 20.0 | 30.0 | 20.0     | 32.0 | 28.0     | 30.0 |
| 1                                                           | 40.0                    | 40.0 | 40.0 | 30.0     | 30.0 | 36.0 | 30.0     | 40.0 | 44.0     | 46.0 |
| 1                                                           | 27.0                    | 28.0 | 30.0 | 21.0     | 30.0 | 34.0 | 26.0     | 31.0 | 30.0     | 30.0 |
| 1                                                           | 20.0                    | 20.0 | 21.0 | 25.0     | 30.0 | 38.0 | 20.0     | 23.0 | 20.0     | 22.0 |
| 1                                                           | 25.0                    | 26.0 | 40.0 | 25.0     | 30.0 | 33.0 | 30.0     | 35.0 | 25.0     | 26.0 |
| 1                                                           | 25.0                    | 27.0 | 28.0 | 30.0     | 36.0 | 39.0 | 20.0     | 25.0 | 20.0     | 28.0 |
| 1                                                           | 25.0                    | 25.0 | 25.0 | 23.0     | 23.0 | 27.0 | 25.0     | 27.0 | 20.0     | 20.0 |
| 1                                                           | 35.0                    | 40.0 | 40.0 | 30.0     | 35.0 | 40.0 | 40.0     | 40.0 | 25.0     | 25.0 |
| 1                                                           | 34.0                    | 35.0 | 35.0 | 30.0     | 39.0 | 39.0 | 37.0     | 39.0 | 35.0     | 35.0 |
| 1                                                           | 20.0                    | 20.0 | 25.0 | 15.0     | 17.0 | 20.0 | 20.0     | 25.0 | 28.0     | 28.0 |
| 1                                                           | 20.0                    | 20.0 | 28.0 | 20.0     | 20.0 | 20.0 | 20.0     | 20.0 | 23.0     | 25.0 |
| 1                                                           | 20.0                    | 25.0 | 30.0 | 15.0     | 15.0 | 15.0 | 30.0     | 35.0 | 30.0     | 35.0 |
| 1                                                           | 25.0                    | 30.0 | 35.0 | 25.0     | 30.0 | 37.0 | 34.0     | 39.0 | 20.0     | 25.0 |
| 1                                                           | 20.0                    | 20.0 | 20.0 | 20.0     | 20.0 | 21.0 | 16.0     | 20.0 | 20.0     | 20.0 |
| 1                                                           | 15.0                    | 20.0 | 24.0 | 16.0     | 20.0 | 20.0 | 15.0     | 18.0 | 15.0     | 15.0 |
| 1                                                           | 35.0                    | 37.0 | 40.0 | 17.0     | 17.0 | 17.0 | 5.0      | 35.0 | 30.0     | 35.0 |
| 1                                                           | 15.0                    | 25.0 | 27.0 | 18.0     | 20.0 | 20.0 | 20.0     | 25.0 | 16.0     | 20.0 |
| 1                                                           | 20.0                    | 26.0 | 28.0 | 20.0     | 20.0 | 25.0 | 10.0     | 23.0 | 10.0     | 25.0 |
| 1                                                           | 20.0                    | 20.0 | 23.0 | 13.0     | 17.0 | 20.0 | 18.0     | 25.0 | 15.0     | 15.0 |
| 1                                                           | R                       | R    | R    | R        | R    | R    | R        | R    | R        | R    |
| 1                                                           | R                       | R    | R    | R        | R    | R    | R        | R    | R        | R    |
| 1                                                           | 30.0                    | 30.0 | 30.0 | 35.0     | 38.0 | 38.0 | 35.0     | 35.0 | 15.0     | 20.0 |
| 1                                                           | 20.0                    | 25.0 | 30.0 | 20.0     | 28.0 | 30.0 | 13.0     | 17.0 | 10.0     | 13.0 |
| 1                                                           | 22.0                    | 38.0 | 40.0 | 37.0     | 40.0 | 45.0 | 40.0     | 40.0 | 30.0     | 40.0 |

ineffective as the organisms are developing resistance to them.

Keeping in view the above introduction different ophthalmic solutions and suspensions of sulfacetamide sodium alone and in combination were evaluated for antibacterial activity against these organisms namely *Escherichia coli* and *Staphylococcus aureus*.

The present study indicated that sulfacetamide alone of four different brands

were evaluated against twenty five clinical isolates of *Escherichia coli*.

Ten (40%) clinical isolates were sensitive to four brands of ophthalmic solutions, fourteen (56%) isolates showed sensitivity against one brand where as one (4%) isolates of *Escherichia coli* are resistant to all preparations Tables 2a and b.

When sulfacetamide was used in combination, Table 3, clinical isolates of

**Table 5**  
Code # of Different Brands used in Combination  
(Sulfacetamide Na 10% + Prednisolone Acetate 0.25%, 0.12 and 0.1%)

| S. No. of clinical isolates of <i>Staphylococcus aureus</i> | Zone of inhibition (mm) |        |        |        |        |        |        |
|-------------------------------------------------------------|-------------------------|--------|--------|--------|--------|--------|--------|
|                                                             | 0.25%                   |        |        |        | 0.12%  |        | 0.1%   |
|                                                             | 05 sol                  | 06 sol | 07 sol | 08 sol | 09 sol | 10 sol | 11 sol |
| 1                                                           | 17.0                    | 16.0   | 20.0   | 17.0   | 17.0   | 20.0   | 10.8   |
| 2                                                           | R                       | R      | 15.0   | 13.0   | R      | R      | R      |
| 3                                                           | 15.0                    | 16.0   | 23.0   | 25.0   | 19.0   | 15.0   | 13.0   |
| 4                                                           | 25.0                    | 23.0   | 28.0   | 26.0   | 25.0   | 26.0   | 22.0   |
| 5                                                           | 23.0                    | 25.0   | 20.0   | 22.0   | 20.0   | 22.0   | 21.0   |
| 6                                                           | 26.0                    | 25.0   | 25.0   | 25.0   | 25.0   | 25.0   | 25.0   |
| 7                                                           | 22.0                    | 20.0   | 20.0   | 20.0   | 21.0   | 20.0   | 20.0   |
| 8                                                           | 21.0                    | 23.0   | 12.0   | 20.0   | 19.0   | 23.0   | 20.0   |
| 9                                                           | 15.0                    | 15.0   | 20.0   | 20.0   | 15.0   | 20.0   | 18.0   |
| 10                                                          | 37.0                    | 37.0   | 33.0   | 35.0   | 37.0   | 35.0   | 35.0   |
| 11                                                          | 25.0                    | 20.0   | 25.0   | 30.0   | 15.0   | 20.0   | 20.0   |
| 12                                                          | 25.0                    | 25.0   | 30.0   | 40.0   | 40.0   | 36.0   | 36.0   |
| 13                                                          | 28.0                    | 27.0   | 25.0   | 30.0   | 25.0   | 25.0   | 18.0   |
| 14                                                          | R                       | R      | R      | R      | 30.0   | 25.0   | R      |
| 15                                                          | 17.0                    | 17.0   | 20.0   | 20.0   | 20.0   | 20.0   | 20.0   |
| 16                                                          | 17.0                    | 17.0   | 18.0   | 12.0   | 20.0   | 10.0   | 10.0   |
| 17                                                          | 20.0                    | 25.0   | 33.0   | 40.0   | 30.0   | 35.0   | 33.0   |
| 18                                                          | 10.0                    | 18.0   | R      | R      | R      | R      | R      |
| 19                                                          | 15.0                    | 16.0   | 14.0   | 17.0   | 25.0   | 20.0   | 10.0   |
| 20                                                          | R                       | R      | R      | R      | R      | 15.0   | R      |
| 21                                                          | R                       | 18.0   | 10.0   | R      | 30.0   | 35.0   | R      |
| 22                                                          | R                       | R      | R      | R      | R      | 15.0   | R      |
| 23                                                          | 25.0                    | 28.0   | 34.0   | 35.0   | R      | R      | 15.0   |
| 24                                                          | R                       | R      | 15.0   | 15.0   | R      | R      | 10.0   |
| 25                                                          | 30.0                    | 32.0   | 36.0   | 34.0   | 32.0   | 30.0   | 34.0   |

*Escherichia coli* have shown that live (20%) isolates were completely sensitive to different brands used while twenty (80%) isolates were found to be resistant to all the brands.

The present studies are in confirmation with reports of Clements *et al* (2002); Hansen (2000); Velschow Bailey *et al* (1998); O'Brien *et al* (1982) and Trolldenier (1995) who reported that sulfacetamide alone as well as in combinations were found to be resistant against *Escherichia coli*.

Antibacterial activity of sulfacetamide alone and all four brands, Table 4, indicated that twenty three (92%) clinical isolates of *Staphylococcus aureus* were sensitive to all the brands used & only two (8%) clinical isolates of *Staphylococcus aureus* were resistant.

Sensitivity of *Staphylococcus aureus* when evaluated against different combinations of sulfacetamide, Table 5, indicated seventeen (68%) isolates were sensitive to all brands while only eight (32%) were resistant to



different combinations.

These results are in confirmation with the reports of Alovero *et al* (2001); Beyer *et al* (2004); Mitchell & Friedlaender (1995); Schumann (1995) and Bailey *et al* (1998), that *Staphylococcus aureus* is sensitive to sulfacetamide alone as well as in combination with trimethoprim and other combinations.

The present study indicated that sulfacetamide alone as well as in combination is not effective against different clinical isolates of *Escherichia coli* while it is still effective against clinical isolates of *Staphylococcus aureus*. Hence the ophthalmic solution and suspension of sulfacetamide and in combinations can be used for the treatment of eye infections caused by *Staphylococcus aureus*, which is very common.

This study will be helpful for different clinicians and pharmacists for prescribing the ophthalmic drug.

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